Ariel University

School of Computer Science

Multimotek

Examining the Impact of a Game on Knowledge and Management of Type 1 Diabetes

Submitted by:
Elizabeth Chepurko

Supervisor:
Prof. Erel Segal-Halevi

July 2026

Acknowledgements

I would like to thanks to my project supervisor, **Prof. Erel Segal-Halevi**, for his invaluable academic guidance and insightful perspective throughout this project. His honest critiques and creative suggestions were instrumental in shaping and improving the game.

A very special thanks goes to **Elinor Benizri**, CEO of "The Center for Diabetes in Israel" (HaMerkaz LeMa'an Cholei Sukaret BeYisrael), and to **Yonatan Izhaki**, a dedicated volunteer partner who supported this project with his technical knowledge in diabetes management and expertise in player experience. I am deeply grateful for their exceptional guidance, planning, and support in distributing the questionnaires and recruiting trial participants. Furthermore, I sincerely appreciate the immense amount of time they both dedicated to multiple meetings and extensive playtests for the game.

Finally, I would like to thank all the participants and testers who generously volunteered their time to evaluate the game. Their valuable feedback and insights were crucial in refining the usability and overall player experience of "Multimotek".

Abstract

Type 1 Diabetes (T1D) is a chronic autoimmune condition requiring continuous, strict daily management, including frequent blood glucose monitoring, precise carbohydrate counting, and timely insulin administration. For newly diagnosed patients and their supporting environments, acquiring the necessary medical knowledge and behavioral adaptation can be overwhelming, as traditional educational materials often lack engagement and fail to sustain long-term motivation. Alongside conventional clinical guidance, there is a growing need for accessible, interactive, and gamified tools capable of delivering essential management strategies while fostering a supportive community.

This project presents **Multimotek**, an interactive game developed in Unity designed to enhance diabetes education, patient engagement, and daily coping mechanisms. Developed on a voluntary basis in close collaboration with *The Center for Diabetes in Israel*, the game's mechanics are systematically aligned with therapeutic needs and medical accuracy to ensure a safe and engaging learning environment. The player progresses through an educational gameplay loop that simulates real-world scenarios, focusing on critical aspects of disease management such as accurate carbohydrate counting, dynamic insulin dosage calculations, and emotional well-being.

Throughout development, playtests were conducted with patients and community members. The study specifically evaluated the game's effectiveness in improving knowledge and daily management skills of Type 1 Diabetes. Initial evaluation results demonstrate a significant increase in diabetes-related knowledge among participants, alongside enhanced confidence in managing daily routines. This suggests that Multimotek effectively translates complex medical protocols into an intuitive and enjoyable digital experience. This report details the game design principles, the technical architecture, and the evaluation outcomes, highlighting the potential of games as innovative tools for chronic illness education.

Contents

1 Introduction

1.1 Background and Prevalence of Type 1 Diabetes

Type 1 Diabetes (T1D) is a chronic, autoimmune metabolic disorder characterized by the immune destruction of insulin-producing beta cells in the pancreas. T1D requires lifelong insulin replacement for survival. This condition impacts millions of individuals globally, with a rising incidence observed in children, adolescents, and young adults [1]. Managing T1D is a 24/7 responsibility that demands absolute precision from both the patients and their immediate support networks, including family members and caregivers.

In the local context, recent data from the Israeli Ministry of Health indicates a profound public health challenge, with nearly 700,000 individuals diagnosed with diabetes nationwide [2]. While Type 2 diabetes represents the vast majority of these cases, Type 1 Diabetes accounts for approximately 10% of the total diabetic population, placing the estimated number of individuals living with T1D in Israel between 50,000 and 60,000 active cases [2, 3].

Importantly, the clinical necessity for intensive insulin management extends beyond T1D. A significant subset of individuals diagnosed with Type 2 diabetes—estimated at roughly 20% to 25% of the T1D-adjacent metabolic spectrum—eventually experience advanced beta-cell exhaustion, transitioning into insulin-dependent Type 2 diabetes [2, 1]. These patients must adopt the exact same daily regimen as T1D patients, including precise carbohydrate counting, continuous glucose monitoring, and dynamic mealtime insulin calculations. Consequently, the actual target demographic requiring immediate behavioral education and simulated training tools like Multimotek is substantially larger, encompassing over 150,000 patients and supporters in Israel alone. The steady increase in these insulin-dependent populations emphasizes the critical need for scalable, accessible educational solutions to help families navigate the complexities of daily disease management.

1.2 Core Management Tasks

Daily care for an individual with T1D revolves around three deeply connected clinical components, as defined by the American Diabetes Association (ADA) guidelines for comprehensive diabetes care [4]:

- **Blood Glucose Regulation:** Patients must frequently monitor their blood sugar levels—utilizing either traditional finger-prick capillary testing or Continuous Glucose Monitors (CGM)—to maintain their glucose levels within a safe target range (typically 70–180 mg/dL). Absolute alertness is required to avoid acute, life-threatening complications such as severe hypoglycemia (critically low blood sugar) or hyperglycemia (caused by profound insulin deficiency and high blood sugar) [4].

- **Carbohydrate Counting:** Nutritional therapy is a cornerstone of daily management. Every meal requires a precise estimation of total carbohydrate content to evaluate its immediate glycemic impact and determine the corresponding metabolic demand [4].
- **Dynamic Insulin Dosing:** Insulin dosages cannot remain static; they must be dynamically calculated for every meal and correction. This calculation relies on personalized clinical parameters, primarily the insulin-to-carbohydrate ratio (ICR), while adjusting for real-time confounding variables such as anticipated physical activity, illness, and acute stress factors [4].

The continuous convergence of these daily tasks creates a severe cognitive and emotional burden. Clinical insights from the *diaTribe Foundation* demonstrate that blood glucose levels are influenced by at least 42 distinct biological, environmental, and behavioral factors—ranging from stress and sleep deprivation to environmental temperature and metabolic timing [5]. Expecting newly diagnosed individuals and their family members to memorize and balance these 42 moving variables through traditional tools creates an exceptionally steep learning curve, frequently leading to management burnout.

To address this overwhelming complexity without causing cognitive overload, Multimotek strategically distills these variables, focusing strictly on the most fundamental core pillars: blood glucose tracking, carbohydrate counting, physical activity, water consumption and insulin administration. By isolating these primary mechanics, the game provides a structured, step-by-step learning environment. This highlights the critical need for interactive, simulated cognitive tools that can simplify multi-variable clinical challenges into a risk-free, and manageable digital experience.

1.3 The Solution

To bridge the gap between passive medical information and active behavioral adaptation, this project introduces **Multimotek**—an interactive, educational game developed in Unity. Designed for individuals with Type 1 Diabetes and their supporting environments. The game features a narrative-driven adventure where players guide a character living with T1D through various daily challenges. By embedding clinical management within an engaging story, the game enhances player immersion, enjoyment, and intrinsic motivation. Within this gameplay loop, players' choices regarding dietary habits, insulin dosing, and daily routine management directly mirror physiological outcomes on the screen. This interactive, character-centric approach allows both patients and caregivers to build intuitive confidence, practice critical decision-making, and navigate realistic diabetes scenarios without real-world medical risk.

1.4 Project Goals and Objectives

The primary goal of this project is to evaluate the impact of a gamified learning environment on the knowledge retention and daily management confidence of T1D patients and their support networks. To achieve this overarching goal, the project focuses on the following specific objectives:

- **Develop a Simulation:** Create an engaging, medically accurate gameplay loop in Unity that models blood glucose tracking, carbohydrate counting, and insulin administration.
- **Promote Advanced Technology Adoption:** Encourage the utilization of insulin pumps by simulating their automated delivery mechanics, demonstrating how these devices simplify blood sugar regulation based on dietary inputs, and reducing patient hesitation toward adopting medical technologies.
- **Engage the Support Ecosystem:** Design the game mechanics to be accessible and educational not only for patients but also for family members and peers, fostering mutual understanding and practical support within the patient's circle.
- **Empower Through Experiential Learning:** Replace dry, passive instructional materials with a practical tool that promotes deep behavioral modification.
- **Reshape Patient Perception:** Utilize the game's interactive format to shift the perception of diabetes management from an overwhelming, overly complex challenge into a manageable daily routine.
- **Conduct Clinical-Community Evaluation:** Perform structured playtests and quantitative evaluations in collaboration with *The Center for Diabetes in Israel* to measure improvements in diabetes knowledge and management.

2 Related Work

The integration of digital interventions in chronic disease management has grown rapidly, particularly for conditions requiring intensive, daily behavioral compliance like Type 1 Diabetes. This section reviews existing digital approaches to diabetes education and highlights the gaps that Multimotek aims to fill.

2.1 Mobile Applications and Digital Health Tools

The current market for digital diabetes management is saturated with mobile health applications, ranging from automated insulin calculators to electronic glucose logbooks. While these tools are highly functional for tracking clinical metrics such as trends in blood glucose levels or active insulin, they are fundamentally designed as utility platforms rather than educational spaces.

Research indicates that traditional health apps often suffer from low long-term user retention and high attrition rates, primarily because they treat patient interactions as monotonous, passive data entry tasks [6]. For newly diagnosed individuals and their families, these tools provide raw computational data but fail to deliver contextual, experiential learning. Furthermore, standard clinical apps frequently assume the user already possesses a high degree of baseline diabetes health literacy, rather than providing the structured, hands-on simulation needed to safely master insulin and carbohydrate calculations.

2.2 Games for Diabetes Education

To counter the dry nature of standard medical software, gamified learning environments and serious games have emerged. Among the limited diabetes-education games currently available, *Level One: A Diabetes Game* represents one of the most relevant and documented examples. An interactive application designed to immerse players in situations requiring blood sugar balance and lifestyle choices. Games of this nature demonstrate the power of virtual environments and making repetitive tasks more engaging through immediate feedback loops.

However, a deeper structural analysis of existing applications like *Level One* reveals core limitations in their pedagogical design and alignment with broader community needs:

- **Didactic "Puzzle" Mechanics vs. Real-World Flow:** Platforms like *Level One* often employ a highly fragmented, didactic "puzzle" approach, breaking down metabolic management into dozens of micro-stages. While this intent aims to simplify learning, it fails to reflect the true, unpredictable lived experience of managing diabetes, which requires continuous multitasking. By forcing users to stop and solve isolated segments, the gameplay becomes overly clinical and fundamentally unengaging, lacking authentic experiential immersion.

- **Individual Isolation:** These games focus almost exclusively on the patient, completely overlooking the critical role of parents, siblings, and friends who form the immediate support ecosystem and share the cognitive management burden.

- **Instruction Fatigue and Restricted Player Agency:** A significant drawback in *Level One* is its heavy reliance on dense, repetitive text instructions to explain core mechanics, which rapidly induces attention fatigue and causes players to disengage. Additionally, its visual framework remains static across different screens, lacking an active narrative to sustain the player's focus. Furthermore, the game strictly restricts player by dictating exactly which food items the character must consume. In contrast, Multimotek combats tutorial fatigue by embedding tasks within an evolving story and granting players complete autonomy to choose any dietary options they desire, realistically simulating the decision-making and dietary freedom required in actual diabetes management.

- **Narrow Target Audience vs. Multi-Stage Adaptability:** As its name implies, *Level One* is strictly tailored for newly diagnosed pediatric patients, focusing exclusively on the immediate, initial introduction to the disease. Consequently, experienced patients or long-term caregivers would find the gameplay irrelevant to their ongoing needs. In contrast, Multimotek offers multi-stage adaptability, framing its narrative and challenges to be valuable for individuals at any phase of their diabetes journey. By presenting scenarios that resonate with both beginners and veteran patients looking to combat management fatigue, the game expands its scope to serve the entire diabetes community regardless of their time since diagnosis.

2.3 Summary of Identified Gaps

In conclusion, the review of current digital diabetes tools reveals a distinct polarization in the digital health market. On one hand, mobile health utilities provide high clinical functionality but fail to deliver engaging, educational experiences for newly diagnosed families. On the other hand, existing games like *Level One* introduce interactive environments but introduce heavy didactic constraints, rigid player choices, tutorial fatigue, and target audience isolation.

Most importantly, none of the reviewed platforms actively target the patient's support ecosystem or incorporate modern management devices, such as insulin pumps, into an experiential learning. This clear divide highlights a critical, unfulfilled need for a localized, narrative-driven digital simulation. A successful framework must balance medical accuracy with high player autonomy, providing an adaptable training ground for both patients and their caregivers. This clear gap in chronic illness education establishes the direct motivation for the development and evaluation of the system presented in this report.

3 System Overview

The design of Multimotek connects three main elements: medical education, interactive gameplay, and technical stability. Instead of keeping learning and playing separate, the game builds Type 1 Diabetes (T1D) management directly into the core mechanics of a 2D platformer.

This section presents a high-level view of how the system works. It covers how the narrative translates real-world medical concepts into gameplay, details the level design and user interface (UI), and explains the technical architecture and data persistence strategies that support the player's experience.

3.1 Narrative Theme and Character Design

The gameplay frames a narrative-driven adventure centered around "Multimotek"—a purple octopus living with T1D. The character's design reflects the official association's symbol, while the octopus itself serves as a metaphor for the continuous multitasking required in diabetes management, where patients must handle multiple physiological responsibilities simultaneously, similar to coordinating eight tentacles. Throughout the journey, the player is accompanied by two diabetic children characters who feature prominently in the instructional interfaces. To foster visual representation, these characters are designed with visible Continuous Glucose Monitor (CGM) sensors attached to their arms, bridging digital storytelling with the lived experiences of patients.

3.2 Structural Level Organization

The game is structurally organized into 8 sequential levels designed to form a progressive pedagogical curve. The framework incrementally introduces environmental challenges and advanced diabetes technologies to guide clinical mastery. A detailed breakdown of each stage's is presented in Section 5.

3.3 Platform Cross-Compatibility

1. **Web Platform (Desktop Layout):** Character navigation (left, right, and climbing) is managed via the keyboard arrow keys. Inventory management, food consumption, and interaction with the digital insulin pump interface are executed using mouse-click inputs.
2. **Mobile Platform (Touch Layout):** The desktop inputs are mapped to a mobile graphical user interface featuring a virtual analog touchpad for multi-directional horizontal movement, combined with a dedicated on-screen jump button.

3.4 Virtual Environment and Level Layout

The game features a 2D world built on platformer mechanics. The level geometry is structurally diverse, requiring players to navigate through several key environmental components:

- **Terrain Variance and Static Hazards:** The levels include vertical geometry such as hills, valleys, and ladders. Embedded within this layout are static hazards like floor spikes, water waves and bottomless pits. Colliding with spikes or falling into pits triggers an immediate state reset, instantly respawning Multimotek at the last activated checkpoint.
- **Embedded Educational Typography:** To integrate learning seamlessly without interrupting gameplay, the terrain itself serves as an instructional canvas. Clinical tips, motivational quotes, and tactical advice are embedded directly into the ground tiles (e.g., reminding players to test rather than guess blood sugar). This approach ensures continuous, non-intrusive learning during exploration.
- **Dynamic Obstacles:** Moving platforms act as horizontal and vertical elevators, requiring timed jumps and spatial awareness.
- **Environmental Entities ("Enemies"):** Hostile entities are placed throughout the map. They act as physiological hazards, triggering immediate increases or decreases in the Multimotek's glucose levels upon collision.

3.5 User Interface (UI) and HUD

The primary objective is to clear each level while maintaining Multimotek's blood glucose within the clinical target range of 70–180 mg/dL. To navigate the world and manage this clinical state, the player utilizes a persistent Heads-Up Display (HUD) positioned on multiple sides of the screen, comprising several critical interface elements:

- **Blood Glucose Monitor Interface:** Displays Multimotek's metabolic state through an evolutionary interface that mirrors real-world disease management:
  - Manual Self-Monitoring (Levels 1–2): Simulating traditional finger-prick blood glucose meters, the monitor remains blank by default. The player must actively trigger an on-screen button to read the current glucose value, which remains visible for a transient window of 3 seconds before fading.
  - Continuous Glucose Monitoring (Levels 3–7): Upon unlocking the CGM technology in Stage 3, the interface transitions into a persistent, real-time display. Mirroring modern clinical sensors.

The underlying pharmacokinetic mechanics driving these continuous backend calculations are detailed in Section 4.2.

- **Translating Metabolic Data to Visual Feedback:**The glucose meter dynamically changes color based on standard real-world medical endocrinology guidelines: Green for target range (70–180 mg/dL), Red for hypoglycemia (less than 70 mg/dL), and Yellow for hyperglycemia (more than 180 mg/dL). To represent the active metabolic derivative, blinking directional arrows flash alongside the meter (yellow pointing up for rising trends, red pointing down for falling trends). Furthermore, an explicit warning icon flashes during critical threshold cases (less than 60 mg/dL or more than 190 mg/dL) to demand immediate intervention.

- **Accelerated Time-of-Day Clock:** Displayed in the bottom left side, this clock visually anchors the player in the game's compressed temporal reality. It allows tracking the passage of time and anticipating metabolic shift.

- **Inventory Subsystem:** A storage interface containing collected food items and insulin syringes, which can be dynamically opened to administer treatments at any time based on player autonomy.

- **Info Button:** A readily accessible UI button that opens a digital guidebook. This reference tool empowers players to independently verify the clinical properties of food assets and enemies.

3.6 Gameplay Flow

1. **Pre-Game Onboarding and Instruction System:** Before entering active gameplay, static instructional slides visually map out the persistent HUD. Prior to the start of every individual level, a dialogue system triggers, where the two diabetic children characters appear on screen to deliver adaptive, level-specific instructions regarding newly unlocked medical devices.

2. **Exploration and Tool Utilization:** The player explores the single-player 2D world, collecting resources and avoiding enemies. In earlier stages, the baseline CGM guides their choices. In later levels (Stages 5–7), an advanced insulin pump interface is added, allowing players to input carbohydrates and experience automated insulin delivery.

3. **The Hidden Item Quest:** To activate the level's exit door, the player must locate a unique hidden artifact concealed within the environment. If the player fails to locate the object after a predetermined threshold, a guidance script instantiates a yellow directional arrow above the character.

4. **Objective and State Continuity:** Upon retrieving the artifact, the player navigates to the exit portal. The session transitions to a post-level statistics panel, freezing the final physiological metrics for evaluation. A detailed breakdown of how these metrics impact level completion and progression is presented in Section 5.

3.7 Technical Architecture and Tools

- **Development Engine (Unity):** Developed using the Unity game engine, leveraging its 2D physics engine, tilemap editors, and UI Toolkit to construct the platformer environments and real-time HUDs.
- **Programming Language and Scripting:** Game logic, physiological state calculations, and UI data-binding were coded in C# using Object-Oriented Programming (OOP) principles and a modular component-based architecture.
- **Source Control and Platform Branching (Git):** Project source code was systematically managed using Git and hosted on GitHub. To develop for both WebGL and Android formats simultaneously without configuration conflicts, a dedicated branching strategy was implemented.

3.7.1 Database Architecture and State Persistence Strategy

The system's backend architecture systematically divides data persistence into three operational categories based on life-cycle, application state, and tactical resets:

- **Persistent Local Storage (PlayerPrefs):** Overarching game progression data, such as level unlock states, are written directly to the local application cache using Unity's `PlayerPrefs` utility class. This ensures data is retained in browser or app sessions, preventing unauthorized resets of progression upon app reloads.
- **Inter-Level State Continuity (Inventory and Glucose):** To simulate the uninterrupted reality of chronic illness management, dynamic parameters must survive scene transitions. Core backend manager scripts use the Singleton design pattern to preserve the player's exact blood glucose value, current time-of-day timestamp, and collected inventory items (food and syringes) between levels, ensuring the player begins a subsequent stage with their exact previous resources and metabolic state.
- **Isolated Level Feedback and Global Aggregation:** During gameplay, a dedicated manager script records the exact time the player spends within the glucose target range. When the level ends, this data is sent to the UI, and the manager resets completely so the next level starts with a clean slate. While local data resets each stage, a global script saves a copy of every level's scores. In the final stage (Level 8), the system combines these saved metrics to calculate and display the player's overall average across the entire game.

4 Diabetes Simulation Engine

4.1 Physiological State Engine

Multimotek enforces illness continuity through two synchronized backend mechanisms:

- **Mathematical Time Compression:** To demonstrate how metabolic shifts compound over hours rather than minutes, the system utilizes the accelerated time-of-day clock described in Section 3.5. Operating at a calculated ratio of 3 game-world hours to 1 real-world minute, this compression allows delayed pharmacokinetic effects—such as insulin degradation and carbohydrate absorption active over hours—to manifest visually within a brief gameplay session.
- **Physiological Parameter Continuity:** The state machine actively preserves metabolic metrics across scene changes. For example, if a player completes Level 1 at 13:05 with a blood glucose level of 190 mg/dL, the simulation engine enforces these precise numbers as the initialization vector for Level 2. Conversely, launching a stage directly from the main menu bypasses this continuity, initializing the engine to a standard fasting baseline at 07:00 AM with a blood glucose of 100 mg/dL.

4.2 Metabolic Superposition and Pharmacokinetic Modeling

A primary technical challenge in developing an authentic T1D simulation is accurately modeling the delayed physiological responses to carbohydrate intake and insulin administration. Multimotek employs a dynamic pharmacokinetic modeling system, internally architected as the Superposition Engine, to calculate blood glucose fluctuations over time based on real-world metabolic absorption rates.

4.2.1 The Superposition Algorithm

Instead of applying flat instantaneous values, every dietary or medical action is instantiated as a time-bound mathematical vector, defined by three core parameters: total glycemic impact (ΔS), onset delay (t_{delay}), and active biological duration (t_{duration}).

Upon consuming an item, the backend engine calculates its specific rate of change r_i (measured in mg/dL per in-game minute):

$$r_i = \frac{\Delta S}{t_{\text{duration}} - t_{\text{delay}}}$$

During active gameplay, multiple metabolic effects frequently overlap (e.g., the player consumes an apple while a previously injected insulin dose is still active). The engine evaluates this continuously by iterating through an active effect queue at every frame-delta. The net metabolic rate $R_{\text{net}}(t)$ applied to Multimotek is calculated using the mathematical principle of superposition:

$$R_{\text{net}}(t) = R_{\text{basal}} + \sum_{i=1}^{n} r_i(t)$$

where R_{basal} represents a continuous background decrease simulating long-acting basal insulin (active primarily in early levels), and n represents the number of concurrently active physiological effects at time t.

4.2.2 Glycemic Index and Absorption Mapping

To ensure clinical authenticity, the parameters assigned to the in-game inventory items strictly reflect their real-world Glycemic Index and biological absorption profiles:

- **Rapid-Acting Carbohydrates (e.g., Cup Of Chocolate, Energy Drinks):** Modeled with a very short onset delay (15 in-game minutes) and a rapid absorption duration (60 in-game minutes). This generates a steep, short-term spike in glucose levels, ideal for treating immediate hypoglycemic events.
- **Complex Carbohydrates (e.g., Bread, Bananas):** Configured with a longer onset delay (20–30 in-game minutes) and an extended duration (120 in-game minutes), resulting in a gradual, sustained glycemic rise.
- **Rapid-Acting Insulin Bolus:** Programmed with a zero-minute in-game onset but an extended biological active duration of 180 game minutes.
- **Zero-Carb Entities (e.g., Fish, Chicken, Vegetables):** Proteins and vegetables provide zero glycemic spikes, effectively teaching the concept of "free foods" in a diabetic diet.

4.3 Automated Insulin Delivery (AID) Subsystem Logic

To support the technology adaptation curve described in the progression design, Stages 5–7 introduce an advanced automated integration mimicking real-world Automated Insulin Delivery (AID) systems. The system transitions the player from manual-intensive calculations to macro-management through three core engineering mechanisms:

4.3.1 Automated Micro-Dosing and Safety Alerts

The primary objective of the insulin pump is to stabilize Multimotek's blood glucose around a default clinical set-point (configured at 105 mg/dL). When glucose rises above this target, the pump automatically administers small, automated micro-boluses instead of requiring manual injections. These micro-boluses function identically to the game's manual insulin syringes, sharing the exact same active lifespan of

180 game minutes. If glucose levels drop rapidly toward the hypoglycemic zone, the pump acts as a critical safety net: it automatically halts all insulin delivery and triggers a prominent warning pop-up at the top-center of the screen when hitting a threshold of 70 mg/dL. Mirroring real-world insulin pump alerts, this non-intrusive modal immediately warns the player to consume carbohydrates before severe hypoglycemia locks in.

4.3.2 The Temporal Activity Target Subsystem

Stage 6 unlocks the Toggle Target 140 feature. Activating this state dynamically overrides the standard regulatory set-point (target = 105 mg/dL) and recalibrates it to a temporary target of 140 mg/dL. When active, the pump dynamically halts automated insulin suppression vectors below this boundary. Furthermore, to simulate the preventative glucose-raising behaviors built into modern commercial insulin pumps prior to physical exertion, the system programmatically drives a gradual, linear glycemic rise toward the 140 mg/dL threshold over a target span of 60 game minutes. The engine calculates the required physiological rate of change for this specific intervention:

$$R_{\text{rise}} = \frac{140 - G_{\text{current}}}{t_{\text{duration}}}$$

Crucially, because this calculated vector is injected into the core Superposition Engine, the net glycemic velocity remains subject to the concurrent metabolic trends active in the frame queue. If the player experiences a steep, heavy downward trend (e.g., from prior active insulin stacking), the continuous addition of R_{rise} will mitigate but may not immediately reverse the descent.

4.3.3 Meal Reporting, Pre-Bolus, and Infusion Delays

When a player eats a food item from their inventory, the Carb Report Manager opens a pop-up window. The system pauses the game world so the player can focus on the calculation without pressure. Once the player enters the correct amount of carbohydrates, they can select a pre-bolus time delay of 0, 15, 30, or 45 game minutes. The system then calculates the expected rise in blood sugar and schedules the insulin delivery. Players who wish to manage the dosage manually can choose to skip the automated injection entirely.

4.4 Physiological Penalties and Failure States

To accurately simulate the physical and cognitive effects of uncontrolled blood glucose, the system dynamically applies kinematic and visual penalties based on Multimotek's metabolic state. Prolonged exposure to extreme glycemic thresholds triggers structured failure states (Game Over), emphasizing the critical nature of timely medical intervention.

- **Hypoglycemia ($<$ 70 mg/dL):** Immediate cognitive and visual impairment is simulated using Unity's Universal Render Pipeline (URP) Post-Processing stack, which applies a progressive blur effect to the camera framework. Because severe hypoglycemia requires urgent real-world intervention, a strict backend timer is initiated. If the player fails to elevate their glucose within 1.5 in-game hours (30 real-world seconds), the screen triggers a gradual blur effect. The game halts, informing the player that Multimotek has fainted, accompanied by a comforting, empathetic educational message before forcing a return to the Main Menu.
- **Hyperglycemia ($>$ 250 mg/dL):** To simulate high blood sugar, Multimotek's horizontal velocity and maximum jump height are reduced, directly increasing the difficulty of platforming challenges. Additionally, the character's animation state machine transitions to a continuous water-drinking idle animation. This visually represents a primary clinical symptom of hyperglycemia, which persists until glucose levels drop below the 250 mg/dL threshold.
- **Severe Hyperglycemia (500+ mg/dL):** Unlike hypoglycemia, which presents an immediate acute danger, hyper-excursions are treated as chronic hazards. A failure state is only triggered if the player remains at or above the 500 mg/dL maximum clamp for a full 24 in-game hours. Once this threshold is breached, active gameplay halts to simulate extreme physical exhaustion. Similar to the hypoglycemia state, the system displays an empathetic educational message explaining the long-term impacts of sustained high blood sugar before returning the player to the Main Menu.

5 Pedagogical Design, Progression, and Feedback

5.1 Progression and Technology Adaptation Framework

The progression architecture utilizes a counter-intuitive "hard-to-easy" framework designed to encourage technology adoption. Early stages present a steep difficulty curve due to the requirements of manual carbohydrate calculation and multiple daily syringe injections. However, as the player advances through the 8 stages, the system layers complexity by introducing novel environmental challenges while gradually unlocking advanced diabetes technologies—specifically continuous monitors and insulin pumps—which automate treatment delivery and alleviate the manual calculation burden.

5.2 Level Progression and Educational Focus Breakdown

To ensure a structured and empathetic learning experience, the game introduces physiological challenges, environmental concepts, and medical technologies progressively. Table 1 details the progression architecture across the 8 stages, mapping out the tactical shift from manual management to advanced device utilization.

Table 1: Detailed Game Stages and Pedagogical Curve

Level	Stage Title & Narrative Theme	Core Educational & Environmental Focus	Active Technology Integration
1	Introduction to T1D	Basic movement onboarding, HUD layout, and target range guidelines.	Manual Calculations (Syringe Only)
2	Introduction to T1D	Elevated terrain difficulty without new technologies to practice baseline dosing.	Manual Calculations (Syringe Only)
3	Continuous Monitoring	Expanded environment (elevators, pits, valleys) with a new enemy; automatic HUD value.	Continuous Glucose Monitor (CGM) Unlocked
4	Nocturnal Hypoglycemia	"The Dark Stage" simulating waking up with low blood sugar; state initializes at 80 mg/dL.	Continuous Glucose Monitor (CGM) Only
5	Transition to Automation	Daybreak environment; introduction to virtual automated insulin delivery and dosing interface.	Basic Insulin Pump Added
6	Advanced Pump Tuning	Utilization of temporary target tuning features adjusted to 140 mg/dL.	Advanced Pump (Target Adjustments)
7	Glycemic Masterclass	High-density hazard layout with maximum enemy saturation; test of full management mastery.	Advanced Insulin Pump Framework
8	Psychological Bonus	Narrative resolution; former hostile entities become peaceful, delivering motivational statements.	Advanced Insulin Pump Framework

5.3 The Evaluation Framework and Time-in-Range (TIR) Metrics

To reinforce learning and provide meaningful reflection, the end of each level triggers a comprehensive performance feedback panel. Rather than merely grading the player with a traditional score, this system translates clinical data into visual, mathematical, and psychological metrics:

- **Time-in-Range (TIR) Metrics:** The backend computing manager calculates and displays the exact percentage of the session's duration that the player spent within three distinct clinical zones:
  - *Hypoglycemic Range:* The percentage of cumulative time spent below the 70 mg/dL threshold ($t_{\text{below}}/t_{\text{total}} \times 100$).
  - *Target In Range (TIR):* The percentage of cumulative time spent within the optimal 70–180 mg/dL target ($t_{\text{in}}/t_{\text{total}} \times 100$).

* *Hyperglycemic Range:* The percentage of cumulative time spent above the 180 mg/dL threshold ($t_{\text{above}}/t_{\text{total}} \times 100$).

- **Mathematical Heart Evaluation and Emotional Alignment** The conversion of physiological performance into visual rewards is executed through a function using Unity's native rounding utilities. The system maps the continuous Time-in-Range percentage (P_{TIR}) into five discrete emotional heart ratings ($H \in \{0, 1, 2, 3, 4, 5\}$). Based on the backend implementation, where the maximum heart constant ($H_{\max}$) equals 5, the core mathematical logic operates as follows:

$$H = \min\left(\max\left(0, \left\lfloor \frac{P_{\text{TIR}}}{100} \cdot 5 \right\rceil\right), 5\right)$$

where $\lfloor \cdot \rceil$ represents the nearest-integer rounding function. Due to the rounding boundaries of this linear scaling equation, the system automatically discretizes player performance into specific clinical threshold categories, as illustrated in Table 2:

Table 2: Time-in-Range (TIR) Percentage to Heart Reward Mapping

TIR Percentage Range (P_{TIR})	**Raw Calculation**	**Rounded Result**	**Heart Reward and Visual State**
90% – 100%	4.50 – 5.00	5	5 Hearts (Optimal Emotional Well-being)
70% – 89%	3.50 – 4.49	4	4 Hearts (Stable Emotional Well-being)
50% – 69%	2.50 – 3.49	3	3 Hearts (Moderate Burnout Risk)
30% – 49%	1.50 – 2.49	2	2 Hearts (Elevated Distress State)
10% – 29%	0.50 – 1.49	1	1 Heart (Severe Emotional Fatigue)
0% – 9%	0.00 – 0.49	0	0 Hearts (Critical Physiological/Mental State)

Crucially, these calculated hearts do not merely represent generic "game lives"; they are systematically coded to symbolize Multimotek's emotional and psychological well-being. By directly tying glycemic stability to the character's emotional state via this algorithmic framework, the mechanic fosters empathy and highlights the connection between physiological health and mental wellness.

- **Empathetic Motivational Statements** Accompanying the quantitative statistics, the panel dynamically generates supportive statements. These phrases are designed to encourage the player to optimize their strategies in subsequent levels, transforming clinical errors into positive learning opportunities.

6 Development Challenges and Mitigations

The development of Multimotek presented multifaceted challenges across game design, technical implementation, and asset management. This section details the primary obstacles encountered during the game development and the strategies employed to mitigate them

6.1 Methodological and Game Design Challenges

Balancing Pedagogy and Engagement One of the core conceptual difficulties was designing an educational game that effectively delivers critical clinical training without sacrificing engagement. Transforming rigid medical protocols into fluid, interactive gameplay required careful prototyping. The challenge lay in ensuring the game did not feel overly didactic, but rather provided immersive mechanics that inherently incentivized players to master blood glucose management.

Requirement Volatility and Medical Complexity Type 1 Diabetes is an incredibly intricate medical condition with dozens of metabolic variables and physiological constraints. Operating in collaboration with *The Center for Diabetes in Israel*, the baseline gameplay requirements experienced significant volatility during the early development stages. Due to the lack of an initially finalized scope, experimental features—such as a dynamic weather-impact subsystem—were programmatically fully implemented and subsequently deprecated to narrow the educational focus onto critical glucose-insulin dynamics. This iterative restructuring created development friction, requiring substantial code refactoring to align strictly with approved medical guidelines.

Inclusive Interface Design and Visual Accessibility Since chronic diabetes can frequently lead to visual impairments, designing the user experience to be universally accessible and visually readable for all players represented a substantial challenge. Balancing a highly informative HUD alongside critical platforming text, tutorial instructions, and action buttons required careful spatial layout optimization to prevent cognitive and visual overload. This challenge became particularly acute during the engineering of Stage 4 (the darkness level). Early playtesting feedback consistently indicated that the environment was excessively dark, rendering it impassable for users with compromised vision. An extensive calibration process was conducted within the Universal Render Pipeline (URP) light settings. Fine-tuning the global ambient light values allowed the system to strike a precise balance between darkness and strict visual accessibility guidelines.

6.2 Technical and Architectural Challenges

State Persistence and Dynamic Scene Calibration Managing data persistence across Unity scene boundaries introduced complex architectural challenges. The system requires continuous tracking of Multimotek's metabolic states—such as precise blood glucose levels, active insulin action queues, and the global clock—as the player progresses through standard levels. Concurrently, the architecture must handle contradictory behaviors: entering a stage from the main menu requires bypassing persistence to initialize clean baseline parameters, while specific educational scenarios demand mid-game overrides.

For instance, Stage 4 introduces a distinct crisis-management scenario that forces a critical hypoglycemic baseline of 80 mg/dL upon entry while completely suppressing the global timer. Resolving the race conditions, memory management conflicts, and dependency breaks caused by these shifting requirements required a complete overhaul of the state management. The monolithic configuration was replaced with modular state injection wrappers, successfully decoupling core variables and allowing the system to dynamically adapt to level-specific constraints.

Universal Render Pipeline (URP) and Package Integration Implementing the Universal Render Pipeline (URP) introduced severe technical hurdles due to package dependencies and installation conflicts. Upgrading the project to URP required overwriting existing rendering packages, which immediately clashed with other active plugins in the game, causing critical asset corruption and compilation errors. Resolving these pipeline conflicts required a meticulous rebuild of the project's dependency tree.

Once stable, URP was used to build custom visual effects that simulate physiological states. A progressive post-processing vignette and lens-blur shader were mapped to the player's glucose levels to represent cognitive deterioration during hypoglycemia. Additionally, for the crisis-management scenario in Stage 4, URP's 2D global illumination and light masking were utilized to construct an environment of dense darkness, requiring careful optimization to maintain mobile cross-compatibility.

6.3 Resource Constraints and Assets

Visual Asset and Graphics Pipelines A significant operational constraint was the lack of dedicated graphic designers or UI/UX artists, requiring the entire art pipeline to be managed and constructed autonomously. To maintain a cohesive and visually appealing aesthetic, a hybrid asset acquisition strategy was developed. This approach combined manual pixel-art sketching, free asset packages, and generative artificial intelligence tools for textures and UI components. Ensuring visual consistency and a unified art style across these disparate asset streams represented a major design and integration challenge.

Audio Voiceovers and Generative Consistency A similar resource constraint arose regarding the game's audio assets, as there were no dedicated voice actors available to record the onboarding instructions at the start of each level. To provide spoken dialogue for the two children characters (a boy and a girl), Google AI Studio was used to generate the voiceovers. However, relying on generative audio tools introduced noticeable consistency issues. The AI-generated voices often varied in tone, emotional delivery, and inflection between different level instructions. Mitigating these fluctuations required extensive trial-and-error prompting and post-processing adjustments to ensure the characters sounded stable and natural throughout the game.

Domain-Specific Clinical Knowledge At the start of the project, a major challenge was the lack of prior knowledge regarding Type 1 Diabetes (T1D). Understanding how complex medical factors—like carbohydrate absorption, insulin decay, and insulin pump algorithms—work together required deep research. To bridge this gap, *The Center for Diabetes in Israel* provided valuable medical guidelines and continuous support. Translating this clinical data into accurate software and gameplay mechanics formed a significant and time-consuming phase of the development lifecycle.

7 Evaluation and Results

This section evaluates the educational impact and efficacy of Multimotek through a structured pre-test and post-test study conducted with 27 participants across three target groups: T1D patients, their support ecosystem, and the general public. The empirical results demonstrate a significant knowledge gain.

7.1 Evaluation Procedure

The evaluation followed a within-subject pre-test/post-test design, structured as follows:

1. **Pre-Test (Questionnaire A):** Participants completed a baseline questionnaire consisting of multiple-choice questions regarding diabetes management.
2. **Gameplay Session:** Participants played Multimotek and were required to reach at least Level 7.
3. **Post-Test (Questionnaire B):** Participants completed the same knowledge questions, along with a verification question confirming they successfully completed Level 7. Data from participants who did not reach Level 7 were excluded.

7.1.1 Participants and Recruitment

To ensure clinical relevance and a diverse sample, a structured community-based evaluation was conducted in close collaboration with *The Center for Diabetes in Israel* to recruit trial participants.

The participant pipeline and data filtration consisted of the following cohorts:

– **Initial Cohort ($N = 75$):** A total of 75 participants were initially recruited and successfully completed the baseline pre-test (Questionnaire A).
– **Completed Cohort ($n = 27$):** Out of the initial group, 27 participants played at least 7 levels of the game and filled out the post-test evaluation (Questionnaire B). This represents a **Completion Rate of 36%**.
– **Comparative Sample:** To ensure statistical validity, the results section analyzes this paired sample of the 27 users who completed both evaluation phases.

The comparative sample spanned a wide demographic range from adolescents to older adults, and was segmented into three distinct target groups representing the game's core audienc:

1. **T1D Patients:** Individuals actively managing Type 1 Diabetes on a daily basis.
2. **The Support Ecosystem:** Family members, partners, peers, and caregivers who share the cognitive and emotional burden of the patient's care.

3. **General Public / Non-Diabetic:** Individuals with no prior personal or clinical connection to diabetes, testing the game's accessibility and educational value for awareness.

The complete list of multiple-choice questions and answers is provided in Appendix A.

7.2 Results

To evaluate the educational impact of Multimotek, participants completed an identical 10-item knowledge questionnaire before the gameplay session (Pre-Test, Phase A) and immediately following the completion of at least seven sequential levels (Post-Test, Phase B). The scores represent the percentage of correct answers, with the post-test verification question excluded from the calculation.

The quantitative data demonstrated a consistent and measurable increase in diabetes management knowledge across all segmented participant groups. The overall mean knowledge score for the entire comparative sample ($n = 27$) improved significantly from a baseline of 74% to 94% post-gameplay.

Table 3 provides a detailed breakdown of the pre-test and post-test performance, alongside the absolute knowledge gains achieved by each target population.

Table 3: Comparative Analysis of User Knowledge Scores (%)

Participant Segment	**Pre-Test (Phase A)**	**Post-Test (Phase B)**	**Absolute Gain**
T1D Patients	75.8%	94.0%	+18.2%
The Support Ecosystem	74.5%	94.5%	+20.0%
General Public / Non-Diabetic	67.5%	92.5%	+25.0%
Overall Average	**74.0%**	**94.0%**	**+20.0%**

The empirical trends illustrated in Table 3 indicate distinct learning trajectories based on prior clinical exposure:

- **T1D Patients:** This group demonstrated the highest baseline knowledge (75.8%), yet achieved a prominent post-test score of 94.0% after playing, indicating the game's utility in reinforcing and solidifying existing clinical concepts.
- **The Support Ecosystem:** Caregivers and family members exhibited a baseline similar to patients (74.5%) and reached the highest overall post-test score at 94.5%, proving the platform's capacity to transfer heavy cognitive care concepts to non-patients.
- **General Public / Non-Diabetic:** This segment recorded the lowest baseline familiarity (67.5%) but achieved the most substantial educational trajectory, with an absolute gain of +25.0%, culminating in a post-test score of 92.5%.

7.3 Discussion and Insights

The consistent increase across all groups suggests that Multimotek effectively supported the acquisition of diabetes-related knowledge. The substantial gain observed within the general public (+25.0%) highlights the game's accessibility, demonstrating that medical protocols can be successfully translated into intuitive mechanics for users with no prior clinical exposure. Furthermore, empowering the support circle through simulation effectively shifts diabetes management from an individual struggle toward a shared family ecosystem, directly alleviating the long-term cognitive and burnout burden on the primary patient.

Limitations and Future Work: While these initial evaluation results are highly promising, the study is bounded by certain limitations. First, the comparative sample size ($n = 27$) is relatively small, meaning broader generalizations require scaling the trial. Second, the study relied on self-reported completion of Level 7 via an explicit questionnaire verification check.

7.4 Summary and Conclusions

The empirical evaluation of Multimotek underscores its efficacy as an interactive, risk-free educational platform capable of driving significant clinical knowledge gains across diverse user cohorts. By translating complex healthcare protocols into fluid game mechanics, the system successfully mitigated tutorial fatigue and converted raw medical data into intuitive behavioral insights.

Based on the study's findings, several core conclusions can be drawn:

- **Clinical Knowledge Acquisition:** Multimotek proved highly effective in delivering complex clinical knowledge, achieving a major +25.0% score increase among individuals with zero prior medical background.
- **Continuous Learning and Reinforcement:** Even for experienced Type 1 Diabetes (T1D) patients with high baseline familiarity, the game served as a powerful continuous learning tool, advancing their mean scores from 75.8% to 94.0%.
- **Support Circle Empowerment:** Immediate caregivers and peers reached an exceptional post-test knowledge threshold of 94.5%. This demonstrates that dedicated digital simulation can successfully decentralize the cognitive burden of disease management, fostering a supportive, knowledgeable family ecosystem around the patient.

In conclusion, Multimotek establishes a scalable, empathetic, and highly impactful framework for chronic illness education, proving that gamified simulations can substantially enhance clinical literacy and empower both patients and their supportive communities.

8 Conclusions and Future Work

8.1 Project Conclusions

This report presented the design, development, and evaluation of Multimotek, an interactive Unity game that turns complex Type 1 Diabetes (T1D) management protocols into an engaging simulation. By modeling key metabolic functions—such as insulin superposition, carb absorption, and automated insulin delivery (AID) mechanics—the game successfully combines medical accuracy with fun gameplay.

The user study, done in close collaboration with *The Center for Diabetes in Israel*, proved that Multimotek is a highly effective educational tool. It helps newly diagnosed populations learn essential concepts, reinforces knowledge for experienced patients, and empowers caregivers to help reduce patient burnout. Ultimately, this project shows that interactive games can bridge the gap between dry medical text and daily life, providing a scalable and empathetic tool for chronic illness education.

8.2 Future Work

While the current version meets its core educational and technical goals, the project can be expanded in the following ways:

- **Complete Mobile Deployment:** Since mobile gaming is highly popular, bringing the game fully to smartphones will increase its reach and accessibility. While the game already supports touch controls and mobile layouts, future work will focus on optimizing the build size and screen scaling for various Android and iOS devices.
- **Multi-Language Support (Localization):** To expand the game's reach and make diabetes education accessible to more diverse communities, future versions will include full localization. While the current game is in Hebrew, adding support for English, Russian, and Arabic will allow a wider range of patients and families to benefit from the platform.
- **Visual Refinement and Better Animations:** Due to a lack of dedicated artists during development, the visual components were created autonomously and supplemented with free online assets and open-source resources. Future versions will focus on collaborating with professional artists to improve character animations and deliver cleaner pixel art environments to enhance player immersion.
- **Audio Effects and Music:** Future updates will introduce unique background music for each level, alongside reactive sound effects for gameplay actions, such as taking damage or entering a hypoglycemic crisis state.
- **New Levels and Scenarios:** Developing more levels based on daily life challenges (like managing sick days or eating out at restaurants) will expand the game's educational value and keep players engaged over time.

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A Evaluation Questionnaire

This appendix presents the complete 10-item multiple-choice questionnaire utilized in both Phase A (Pre-Test) and Phase B (Post-Test) to evaluate participants' knowledge of Type 1 Diabetes management.

1. **How often is long-acting (basal) insulin typically injected?**

   Twice a day

   Once a day

   Three times a day

   Four times a day

2. **What is the primary effect of rapid-acting (bolus) insulin?**

   Lowering blood glucose levels

   Raising blood glucose levels

   Maintaining glucose levels strictly during meal consumption

   Returning blood glucose levels to pre-meal values instantly

3. **What does the clinical term "Time in Range" (TIR) signify?**

   The amount of time it takes for insulin to begin taking effect post-injection

   The total duration during which blood glucose is too low (hypoglycemia)

   The percentage of time or hours spent within the target glucose range (e.g., 70 to 180 mg/dL)

   The timeframe during which an insulin pump is connected to a charger

4. **How many times a day should blood glucose be checked using a traditional glucometer?**

   Once a day

   Only once in the morning

   Twice a day (morning and evening)

   There is no fixed number of times; it should be checked as needed based on symptoms and routine

5. **When should an individual monitor their blood glucose levels in relation to physical exercise?**

   Exclusively before and after the activity

   There is no clinical necessity to monitor during exercise

   Only if the individual experiences symptoms of discomfort

   Before, during, and after the physical activity

6. **What is a Continuous Glucose Monitor (CGM) / Glucose Sensor?**

   A sensor that reads glucose levels only upon a finger-prick and displays subsequent trends

- A wearable device that automatically tracks glucose levels 24/7 at regular intervals
- A medical device that automatically injects insulin without any user intervention
- A standalone mobile application designed solely to calculate meal carbohydrates

7. **What is an insulin pump?**
   - A device that draws blood to analyze current glucose readings
   - A device that extracts excess insulin from the body in the event of an overdose
   - A small computerized device attached to the body that delivers continuous insulin at a fixed rate (basal) and precise doses for meals (bolus)
   - A smartwatch that tracks heart rate and physical expenditure
8. **What is a smart/hybrid closed-loop insulin pump system?**
   - An automated system that communicates with a CGM sensor to automatically increase, decrease, or suspend insulin delivery based on predicted glucose levels
   - An insulin pump that operates strictly on solar-powered batteries
   - A pump that is physically restricted to delivering only one specific brand of insulin
   - An application-based pump designed to remind the user when it is time to eat
9. **Approximately how many grams of carbohydrates are contained in a medium banana weighing 110 grams?**
   - 25 grams
   - 14 grams
   - 110 grams
   - 52 grams
10. **Select the food group that contains the absolute minimum amount of carbohydrates:**
    - Carrot + Bread + Yogurt + Cucumber
    - Hard candy + Peanut puffs (Bamba) + Fish + Dairy ice cream scoop
    - Cucumber + Carrot + Chicken drumstick + Fish
    - Hamburger + Cake + Milk chocolate + Banana